

Fine-Structure Constant: Unit Conventions Why $\alpha = 1$ can be set

Supplement to Document 011

Contents

.1 Introduction and Reference to Document 011 1

 Delimitation from Document 011 1

 Why Two Documents? 2

.2 Different Unit Conventions for α 2

 Overview of Systems 2

.3 Heaviside-Lorentz Units in Detail 2

 What are Heaviside-Lorentz Units? 2

 Why $4\pi\epsilon_0 = 1$ 3

 Fine-Structure Constant in Heaviside-Lorentz 3

.4 Two Variants of Heaviside-Lorentz 3

 Variant A 3

 Variant B: Keep e , $\alpha \approx 1/137$ 4

 Which variant is used? 4

.5 Reconstruction of SI Values from T0 5

 The Central Principle 5

 Example: Fine-Structure Constant 5

 Example: Gravitational Constant 6

 Why does this work? 6

 Comparison Table 6

 Important Conclusion 6

.6 Why can $\alpha = 1$ be set? 7

 Fundamental Insight 7

 Step-by-Step Justification 7

 Analogy: Temperature Scales 8

.7 Consequences of Redefining the Coulomb 8

 What does it mean to redefine elementary charge? 8

 Effects on Electromagnetic Quantities 8

 Electric Current (Ampere) 8

 Electric Voltage (Volt) 9

 Capacitance (Farad) 9

 Are these changes "real"? 9

.8	Practical Impacts on Everyday Calculations	9
	Motivation	9
	Example 1: Ohm's Law	9
	In SI Units (Standard)	9
	In Heaviside-Lorentz with $\alpha = 1$	10
	Example 2: Power of a Light Bulb	10
	In SI Units	10
	In Heaviside-Lorentz with $\alpha = 1$	10
	Example 3: Charging a Capacitor	11
	In SI Units	11
	In Heaviside-Lorentz with $\alpha = 1$	11
	Example 4: RC Time Constant	12
	In SI Units	12
	In Heaviside-Lorentz with $\alpha = 1$	12
	Summary of Practical Calculations	13
	Important Insights	13
	Why does nobody use $\alpha = 1$ in practice?	13
.9	Practical Aspects of Different Systems	14
	Advantages and Disadvantages: SI Units	14
	Advantages and Disadvantages: Heaviside-Lorentz with $\alpha = 1$	14
	Advantages and Disadvantages: Natural Units with $\alpha \approx 1/137$	15
.10	Historical Development	15
	Gauss Units (cgs)	15
	SI Units (MKSA)	15
	Heaviside-Lorentz	15
	Natural Units	16
.11	Fine's Inequality vs. Fine-Structure Constant	16
	Frequent Confusion	16
	Fine's Inequality	16
	Fine-Structure Constant	16
	No Connection!	16
.12	Summary	17
	Core Statements	17
	For Further Details See Document 011	17
0.1	Conversion Table: SI $\leftrightarrow$ Heaviside-Lorentz	17
0.2	Sample Calculation: Coulomb's Law	18
	In SI Units	18
	In HL Units ($\alpha = 1$)	18

Abstract

This document addresses aspects of the fine-structure constant not discussed in detail in Document 011. The focus is on providing a comprehensive justification for why and how $\alpha = 1$ can be set (Heaviside-Lorentz convention), the physical consequences of different unit systems, and the historical and practical implications of redefining electromagnetic units.

For T0-specific derivations (characteristic energy E_0 , geometric parameter ξ , T0 formula $\alpha = \xi(E_0/1\text{MeV})^2$) see Document 011.

1 Introduction and Reference to Document 011

Delimitation from Document 011

Document 011 covers in detail:

- T0 derivation: $\alpha = \xi(E_0/1\text{MeV})^2$
- Characteristic energy: $E_0 = \sqrt{m_e \cdot m_\mu} = 7.398 \text{ MeV}$
- Geometric parameter: $\xi = \frac{4}{3} \times 10^{-4}$
- Alternative formulations: with μ_0 , with r_e/λ_C , etc.
- Historical context (Sommerfeld)
- Natural units and energy as fundamental field
- Detailed dimensional analysis of all formulations

This document (044) focuses on:

- **Why** $\alpha = 1$ can be set (detailed justification)
- **How** different unit conventions work
- Consequences of redefining the Coulomb
- Heaviside-Lorentz vs. Gauss vs. SI units
- Practical aspects and historical development
- Fine's inequality vs. fine-structure constant (name confusion)

Why Two Documents?

Document 011: T0 theory and physical derivations

Document 044: Unit systems and conventions

Both complement each other, with minimal overlap.

.2 Different Unit Conventions for α

Overview of Systems

The fine-structure constant can be expressed in different unit systems:

System	Formula	Value
SI Standard	$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c}$	$\approx \frac{1}{137}$
Heaviside-Lorentz	$\alpha = \frac{e^2}{4\pi}$ (with $\hbar = c = 4\pi\epsilon_0 = 1$)	1 or $\frac{1}{137}$
Gauss (cgs)	$\alpha = \frac{e^2}{\hbar c}$	$\approx \frac{1}{137}$

Table 1: Unit systems for α

Important: The numerical value depends on the convention, the *physical* predictions do not!

.3 Heaviside-Lorentz Units in Detail

What are Heaviside-Lorentz Units?

The Heaviside-Lorentz system is a variant of natural units, specifically for electrodynamics:

$$\boxed{\hbar = c = 4\pi\epsilon_0 = 1} \quad (1)$$

Consequences:

- Electromagnetic equations become more symmetric
- The factor 4π disappears from many formulas
- Elementary charge is redefined

Why $4\pi\epsilon_0 = 1$?

In SI units, 4π appears in many electromagnetic formulas:

$$\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r} \quad (\text{Coulomb's law}) \quad (2)$$

$$\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{Maxwell's equation}) \quad (3)$$

With $4\pi\epsilon_0 = 1$, these become:

$$\vec{E} = \frac{q}{r^2} \hat{r} \quad (4)$$

$$\nabla \cdot \vec{E} = 4\pi\rho \quad (5)$$

The factor 4π moves from Coulomb's law to Poisson's equation!

Fine-Structure Constant in Heaviside-Lorentz

Starting point (SI):

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (6)$$

With $\hbar = c = 4\pi\epsilon_0 = 1$:

$$\alpha = \frac{e^2}{1 \cdot 1 \cdot 1} = e^2 \quad (7)$$

Now the crucial question: What value does e have in this system?

.4 Two Variants of Heaviside-Lorentz

Variant A: Normalize e so that $\alpha = 1$

Approach: We define the unit of charge such that $\alpha = 1$.

Since $\alpha = e^2$ in HL units:

$$e^2 = 1 \quad \Rightarrow \quad e = 1 \quad (8)$$

Physical meaning:

- Elementary charge becomes a *dimensionless unit*
- Electromagnetic coupling is "normalized"
- Charge is measured in units of $\sqrt{\hbar c}$

What changes?

The elementary charge gets a new numerical value:

$$e_{\text{HL}} = \sqrt{4\pi\epsilon_0\hbar c} \quad (\text{expressed in SI units}) \quad (9)$$

Numerically:

$$e_{\text{HL}} = \sqrt{4\pi \times 8.854 \times 10^{-12} \times 1.055 \times 10^{-34} \times 3 \times 10^8} \quad (10)$$

$$\approx 5.29 \times 10^{-19} \quad (\text{new charge unit}) \quad (11)$$

This is about $\sqrt{137} \times e_{\text{SI}}$!

Variant B: Keep e , $\alpha \approx 1/137$

Approach: The elementary charge retains its "natural" value.

In this case:

$$\alpha = e^2 \approx \frac{1}{137} \quad (12)$$

because e in these units has the value $\approx 1/\sqrt{137}$.

Physical meaning:

- Charge retains physical meaning
- α remains $\approx 1/137$
- Only the mathematical form simplifies

Which variant is used?

In practice:

- **T0 theory:** Sets ****all**** constants = 1 ($c = \hbar = \alpha = G = 1$)
- **Theoretical high-energy physics:** Often $\hbar = c = 1$, sometimes also $\alpha = 1$
- **Numerical calculations:** Often $\hbar = c = 1$, but $\alpha \approx 1/137$
- **Experimental physics:** Almost always SI units (all constants have numerical values)

T0 convention:

- In T0 calculations: $c = \hbar = \alpha = G = 1$ (maximum simplification)
- Only free parameter: $\xi = \frac{4}{3} \times 10^{-4}$
- When comparing with experiments: SI values ($c = 3 \times 10^8$ m/s, $\alpha \approx 1/137$, etc.)
- Both describe the same physics!

.5 Reconstruction of SI Values from T0

The Central Principle

Important insight: Although T0 sets all constants to 1, the SI values can be reconstructed!

T0 Reconstruction

In T0 calculations:

- All constants = 1: $c = \hbar = \alpha = G = 1$
- Only free parameter: $\xi = \frac{4}{3} \times 10^{-4}$
- Formulas maximally simplified

Reconstruction of SI values:

- Fine-structure constant: $\alpha_{\text{SI}} = \xi(E_0/1\text{MeV})^2 \approx 1/137$
- Gravitational constant: $G_{\text{SI}} = \frac{\xi^2}{4m_e} \times \text{factors}$
- All other constants: derivable from ξ

Example: Fine-Structure Constant

In T0 units:

$$\alpha = 1 \quad (13)$$

Reconstruction of SI value:

$$\alpha_{\text{SI}} = \xi \cdot \left(\frac{E_0}{1\text{MeV}} \right)^2 \quad (14)$$

With $\xi = \frac{4}{3} \times 10^{-4}$ and $E_0 = 7.398\text{ MeV}$:

$$\alpha_{\text{SI}} = 1.3333 \times 10^{-4} \times (7.398)^2 \quad (15)$$

$$= 1.3333 \times 10^{-4} \times 54.73 \quad (16)$$

$$= 7.297 \times 10^{-3} \quad (17)$$

$$= \frac{1}{137.04} \quad (18)$$

$$\text{Experimental: } \alpha_{\text{exp}} = \frac{1}{137.036}$$

Agreement: 0.03% ✓

Example: Gravitational Constant

In T0 units:

$$G = 1 \quad (19)$$

Reconstruction of SI value:

$$G_{\text{SI}} = \frac{\xi^2}{4m_e} \times C_{\text{dim}} \times C_{\text{conv}} \quad (20)$$

where:

- C_{dim} = Dimension conversion (natural units $\rightarrow$ SI)
 - C_{conv} = Conversion factors (eV $\rightarrow$ J, etc.)
- Detailed derivation see Document 012 (Gravitation).

Why does this work?

Key: ξ is dimensionless and universal!

1. In T0: ξ determines all coupling strengths
2. In SI: ξ together with characteristic energies (E_0 , masses) reconstructs all constants
3. Physical predictions: identical in both systems!
4. Only the mathematical representation differs

Comparison Table

Constant	T0	SI	Reconstruction
c	1	$3 \times 10^8 \text{ m/s}$	Convention
$\hbar$	1	$1.055 \times 10^{-34} \text{ J}\cdot\text{s}$	Convention
α	1	$\approx 1/137$	$\xi(E_0/1\text{MeV})^2$
G	1	$6.67 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$	$\xi^2/(4m_e)\times \text{factors}$
ξ	$\frac{4}{3} \times 10^{-4}$	$\frac{4}{3} \times 10^{-4}$	Same!

Table 2: T0 vs. SI - Reconstruction of constants

Important Conclusion

- **T0 is not new physics**, but a reparametrization
- Instead of many constants ($c, \hbar, \alpha, G, \dots$) only **one parameter** ξ
- All SI values reconstructable from ξ and energy scales
- Advantage: Formulas simpler, physical relationships clearer
- Disadvantage: Conversion to SI needed for experiments

.6 Why can $\alpha = 1$ be set?

Fundamental Insight

Core Statement

The fine-structure constant α is a **dimensionless number**. Its numerical value is **convention-dependent**, not fundamental! One can set $\alpha = 1$ by redefining the **unit of charge** accordingly.

Step-by-Step Justification

Step 1: What is a convention?

SI units are historically evolved definitions:

- 1 meter = originally 1/10,000,000 of the Earth's meridian
- 1 second = originally 1/86,400 of a solar day
- 1 Coulomb = defined via Ampere and force between currents

None of these is "fundamental"!

Step 2: α in SI

In SI units:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \approx \frac{1}{137} \quad (21)$$

The value $1/137$ follows from:

- How we defined the Coulomb (historically)
- How we defined ϵ_0 (via μ_0 and c)

Step 3: Redefinition

We can say: "From now on, the elementary charge is no longer 1.602×10^{-19} C, but $e = \sqrt{4\pi\epsilon_0\hbar c}$."

Then automatically:

$$\alpha = \frac{(\sqrt{4\pi\epsilon_0\hbar c})^2}{4\pi\epsilon_0\hbar c} = 1 \quad (22)$$

Step 4: Physical Consequences

- **No physical predictions change!**
- Only the *numbers* in formulas change
- All ratios remain the same
- All experiments yield the same results

Analogy: Temperature Scales

Celsius: Water freezes at 0°C

Fahrenheit: Water freezes at 32°F

Kelvin: Water freezes at 273.15 K

Is any of these scales "correct"? No! They are conventions.

Similarly, $\alpha = 1/137$ (SI) vs. $\alpha = 1$ (HL) is just a choice of convention!

.7 Consequences of Redefining the Coulomb

What does it mean to redefine elementary charge?

If e is redefined such that $\alpha = 1$:

Old definition (SI):

$$e = 1.602 \times 10^{-19} \text{ C} \quad (23)$$

New definition (HL with $\alpha = 1$):

$$e = 1 \quad (\text{dimensionless in natural units}) \quad (24)$$

or expressed in SI units:

$$e_{\text{new}} = \sqrt{4\pi\epsilon_0\hbar c} \approx 5.29 \times 10^{-19} \text{ (new charge unit)} \quad (25)$$

Effects on Electromagnetic Quantities

Electric Current (Ampere)

Since $1 \text{ A} = 1 \text{ C/s}$:

$$1 \text{ A}_{\text{new}} = \frac{e_{\text{new}}}{1 \text{ s}} = \sqrt{137} \times 1 \text{ A}_{\text{old}} \quad (26)$$

Electric Voltage (Volt)

$1 \text{ V} = 1 \text{ J/C}$:

$$1 \text{ V}_{\text{new}} = \frac{1 \text{ J}}{e_{\text{new}}} = \frac{1}{\sqrt{137}} \times 1 \text{ V}_{\text{old}} \quad (27)$$

Capacitance (Farad)

$$1 \text{ F}_{\text{new}} = \frac{e_{\text{new}}}{1 \text{ V}_{\text{new}}} = 137 \times 1 \text{ F}_{\text{old}} \quad (28)$$

Are these changes "real"?

No! They are only conversion factors, like Celsius $\rightarrow$ Fahrenheit.

All physical ratios remain identical:

- Capacitance of a capacitor / distance: same
- Force between charges / distance²: same
- All experiments: same results

Only the *numerical values* we calculate with change!

.8 Practical Impacts on Everyday Calculations

Motivation

Question: If we set $\alpha = 1$, what does that mean for ordinary electrical calculations with volts, amperes, resistance, capacitance?

Answer: All formulas change, but the *physical results* remain identical!

Example 1: Ohm's Law

In SI Units (Standard)

$$U = R \cdot I \quad (29)$$

Numerical example:

- Resistance: $R = 100 \, \Omega$
- Current: $I = 2 \, \text{A}$
- Voltage: $U = 100 \times 2 = 200 \, \text{V}$

In Heaviside-Lorentz with $\alpha = 1$

The formula remains $U = R \cdot I$, but the *numerical values* change!

Unit conversion:

$$1 \, \text{A}_{\text{new}} = \sqrt{137} \times 1 \, \text{A}_{\text{old}} \approx 11.7 \, \text{A}_{\text{old}} \quad (30)$$

$$1 \, \text{V}_{\text{new}} = \frac{1}{\sqrt{137}} \times 1 \, \text{V}_{\text{old}} \approx 0.085 \, \text{V}_{\text{old}} \quad (31)$$

$$1 \, \Omega_{\text{new}} = \frac{1}{137} \times 1 \, \Omega_{\text{old}} \quad (32)$$

Same circuit in new units:

$$R_{\text{new}} = 100 \times \frac{1}{137} \approx 0.73 \, \Omega_{\text{new}} \quad (33)$$

$$I_{\text{new}} = 2 \times \sqrt{137} \approx 23.4 \, \text{A}_{\text{new}} \quad (34)$$

$$U_{\text{new}} = 0.73 \times 23.4 = 17.1 \, \text{V}_{\text{new}} \quad (35)$$

Conversion back to SI:

$$U_{\text{new}} = 17.1 \times 0.085 \, \text{V}_{\text{old}} = 200 \, \text{V} \quad (36)$$

Identical result!

Example 2: Power of a Light Bulb**In SI Units**

$$P = U \cdot I = \frac{U^2}{R} \quad (37)$$

Light bulb: 60 W at 230 V

$$R = \frac{U^2}{P} = \frac{(230)^2}{60} = 882 \, \Omega \quad (38)$$

$$I = \frac{P}{U} = \frac{60}{230} = 0.26 \, \text{A} \quad (39)$$

In Heaviside-Lorentz with $\alpha = 1$

Power: $1 \, \text{W}_{\text{new}} = 1 \, \text{W}_{\text{old}}$ (energy/time doesn't change in HL!)

$$U_{\text{new}} = 230 \times \frac{1}{\sqrt{137}} = 19.6 \, \text{V}_{\text{new}} \quad (40)$$

$$R_{\text{new}} = 882 \times \frac{1}{137} = 6.44 \, \Omega_{\text{new}} \quad (41)$$

$$I_{\text{new}} = \frac{P}{U_{\text{new}}} = \frac{60}{19.6} = 3.06 \, \text{A}_{\text{new}} \quad (42)$$

Verification:

$$P = U_{\text{new}} \cdot I_{\text{new}} = 19.6 \times 3.06 = 60 \, \text{W} \quad (43)$$

Example 3: Charging a Capacitor

In SI Units

$$Q = C \cdot U \quad (44)$$

Capacitor: $C = 100 \mu\text{F}$ at $U = 12 \text{ V}$

$$Q = 100 \times 10^{-6} \times 12 = 1.2 \times 10^{-3} \text{ C} \quad (45)$$

Stored energy:

$$E = \frac{1}{2}CU^2 = \frac{1}{2} \times 100 \times 10^{-6} \times 144 = 7.2 \times 10^{-3} \text{ J} \quad (46)$$

In Heaviside-Lorentz with $\alpha = 1$

Conversion:

$$1 \text{ F}_{\text{new}} = 137 \times 1 \text{ F}_{\text{old}} \quad (47)$$

$$1 \text{ C}_{\text{new}} = \sqrt{137} \times 1 \text{ C}_{\text{old}} \quad (48)$$

$$C_{\text{new}} = 100 \times 10^{-6} \times 137 = 0.0137 \text{ F}_{\text{new}} \quad (49)$$

$$U_{\text{new}} = 12 \times \frac{1}{\sqrt{137}} = 1.025 \text{ V}_{\text{new}} \quad (50)$$

$$Q_{\text{new}} = 0.0137 \times 1.025 = 0.014 \text{ C}_{\text{new}} \quad (51)$$

Conversion back:

$$Q_{\text{new}} = 0.014 \times \frac{1}{\sqrt{137}} = 1.2 \times 10^{-3} \text{ C}_{\text{old}} \quad (52)$$

Energy:

$$E_{\text{new}} = \frac{1}{2} \times 0.0137 \times (1.025)^2 = 7.2 \times 10^{-3} \text{ J} \quad (53)$$

Energy is the same in all systems!

Example 4: RC Time Constant

In SI Units

$$\tau = R \cdot C \quad (54)$$

Circuit: $R = 1 \text{ k}\Omega$, $C = 10 \mu\text{F}$

$$\tau = 1000 \times 10 \times 10^{-6} = 0.01 \text{ s} = 10 \text{ ms} \quad (55)$$

In Heaviside-Lorentz with $\alpha = 1$

$$R_{\text{new}} = 1000 \times \frac{1}{137} = 7.3 \, \Omega_{\text{new}} \quad (56)$$

$$C_{\text{new}} = 10 \times 10^{-6} \times 137 = 1.37 \times 10^{-3} \, \text{F}_{\text{new}} \quad (57)$$

$$\tau_{\text{new}} = 7.3 \times 1.37 \times 10^{-3} = 0.01 \, \text{s} = 10 \, \text{ms} \quad (58)$$

Time remains the same! This is important: Physical timescales do not change!

Summary of Practical Calculations

Quantity	SI	HL Factor	HL ($\alpha = 1$)
Charge (Q)	C	$\sqrt{137}$	$\sqrt{137} \, \text{C}$
Current (I)	A	$\sqrt{137}$	$\sqrt{137} \, \text{A}$
Voltage (U)	V	$1/\sqrt{137}$	$\text{V}/\sqrt{137}$
Resistance (R)	Ω	$1/137$	$\Omega/137$
Capacitance (C)	F	137	137 F
Power (P)	W	1	W (unchanged!)
Energy (E)	J	1	J (unchanged!)
Time (τ)	s	1	s (unchanged!)

Table 3: Conversion factors SI $\rightarrow$ HL with $\alpha = 1$

Important Insights

Core Statement

What changes:

- Numerical values for charge, current, voltage, resistance, capacitance

What does NOT change:

- Energy
- Power
- Time
- All physical ratios
- All experimental results

Conclusion: It's just a conversion, like meters $\leftrightarrow$ feet!

Why does nobody use $\alpha = 1$ in practice?

Reasons:

1. **Measuring devices:** All voltmeters, ammeters, etc. are calibrated in SI
2. **Standards:** Worldwide accepted SI definitions
3. **Intuition:** Engineers know typical values in SI
 - Household: 230 V, not $1.96 V_{\text{new}}$
 - USB: 5 V, not $0.43 V_{\text{new}}$
4. **Conversion is laborious:** $\sqrt{137}$ factors everywhere
5. **No advantage for practitioners:** Simplification only visible in theoretical formulas

But: For theoretical calculations (QED, Feynman diagrams), $\alpha = 1$ is often very helpful!

.9 Practical Aspects of Different Systems

Advantages and Disadvantages: SI Units

Advantages:

- Worldwide standardized

- Directly usable for experiments
- All measuring devices calibrated in SI
- Clear separation of length/time/mass/charge

Disadvantages:

- Many constants in formulas ($4\pi\epsilon_0$, $\hbar$, c)
- Physical relationships obscured
- Dimensions unwieldy

Advantages and Disadvantages: Heaviside-Lorentz with $\alpha = 1$

Advantages:

- Maximally simplified formulas
- Electromagnetic symmetry visible
- Theoretical calculations simpler
- QED Feynman diagrams more elegant

Disadvantages:

- No direct connection to experiments
- Conversion to SI laborious
- Unfamiliar for practitioners
- Physical "size" of e unclear

Advantages and Disadvantages: Natural Units with $\alpha \approx 1/137$

Advantages:

- Simplified formulas ($\hbar = c = 1$)
- α retains physical meaning
- Good compromise theory/practice
- Numerically: $\alpha \ll 1 \rightarrow$ perturbation theory

Disadvantages:

- Still need conversion to SI
- Factor 4π remains in some formulas

This is the preferred convention in modern particle physics!

.10 Historical Development

Gauss Units (cgs)

19th century: Gauss system (centimeter-gram-second)

$$\alpha = \frac{e^2}{\hbar c} \quad (59)$$

No $4\pi\epsilon_0$, because $\epsilon_0 = 1$ by definition in cgs!

SI Units (MKSA)

20th century: SI system (meter-kilogram-second-ampere)

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (60)$$

The $4\pi\epsilon_0$ appears because the SI ampere is defined via force.

Heaviside-Lorentz

Theoretical physics: Heaviside-Lorentz simplifies Maxwell's equations

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}, \quad \nabla \times \vec{B} = \vec{j} + \frac{\partial \vec{E}}{\partial t} \quad (61)$$

Symmetric! (In SI, μ_0 and ϵ_0 appear asymmetrically)

Natural Units

Modern high-energy physics: $\hbar = c = 1$, but different conventions for α

.11 Fine's Inequality vs. Fine-Structure Constant

Frequent Confusion

Warning: *Fine's inequality* and the *fine-structure constant* are completely different concepts!

Fine's Inequality

What it is:

- A form of Bell's inequality
- Test for local hidden variables
- Quantum entanglement vs. classical correlations

Mathematically:

$$|C(\alpha, \beta) - C(\alpha, \beta')| + |C(\alpha', \beta) + C(\alpha', \beta')| \leq 2 \quad (62)$$

where C are correlation functions.

Physically: Shows non-locality of quantum mechanics

Fine-Structure Constant

What it is:

- Fundamental physical constant
- Strength of the electromagnetic interaction
- Dimensionless, $\alpha \approx 1/137$

Mathematically:

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} \quad (63)$$

Physically: Determines EM coupling strength

No Connection!

The similarity in name is **pure coincidence**. The two concepts have nothing to do with each other!

.12 Summary

Core Statements

1. α is dimensionless $\rightarrow$ numerical value is convention-dependent
2. One **can** set $\alpha = 1$ by redefining the unit of charge
3. **T0 theory:** Sets ****all**** constants = 1: $c = \hbar = \alpha = G = 1$
4. Only free parameter in T0: $\xi = \frac{4}{3} \times 10^{-4}$

5. **No** physical predictions change!
6. Only numerical values in formulas differ
7. When comparing with experiments: SI values ($\alpha \approx 1/137$, $c = 3 \times 10^8$ m/s, etc.)

For Further Details See Document 011

- T0 derivation of α
- Characteristic energy E_0
- Geometric parameter ξ
- Experimental verification
- Detailed dimensional analysis
- Historical context (Sommerfeld)

0.1 Conversion Table: SI $\leftrightarrow$ Heaviside-Lorentz

Quantity	SI	HL ($\alpha = 1$)
Elementary charge	$e = 1.602 \times 10^{-19}$ C	$e = 1$
Fine-structure constant	$\alpha \approx 1/137$	$\alpha = 1$
$4\pi\epsilon_0$	1.11×10^{-10} F/m	1
$\hbar$	1.055×10^{-34} J·s	1
c	3×10^8 m/s	1

Table 4: Conversion table SI to HL

0.2 Sample Calculation: Coulomb's Law

In SI Units

$$F = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2} \quad (64)$$

Numerically for $r = 1 \text{ \AA} = 10^{-10} \text{ m}$:

$$F = \frac{1}{4\pi \times 8.854 \times 10^{-12}} \frac{(1.602 \times 10^{-19})^2}{(10^{-10})^2} \quad (65)$$

$$\approx 2.3 \times 10^{-8} \text{ N} \quad (66)$$

In HL Units ($\alpha = 1$)

$$F = \frac{e^2}{r^2} = \frac{1}{r^2} \quad (67)$$

With r in natural units: $r = 1 \text{ \AA} = 0.197 \times 10^6 \text{ eV}^{-1}$

$$F = \frac{1}{(0.197 \times 10^6)^2} \approx 2.6 \times 10^{-14} \text{ eV}^2 \quad (68)$$

Conversion to SI: $1 \text{ eV}^2 \approx 9 \times 10^5 \text{ N}$

$$F \approx 2.3 \times 10^{-8} \text{ N} \quad (69)$$

Identical! Only the intermediate steps look different.